

MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 7

Modulus

- Q1** Find the range of x where $||x - 5| - 6| > 5$?
 (A) $(-\infty, 6)$ or $(16, \infty)$
 (B) $(-\infty, -1) \cup (4, 6) \cup (7, 16)$
 (C) $(-5, 6) \cup (6, \infty)$
 (D) $(-\infty, -6) \cup (4, 6) \cup (16, \infty)$
- Q2** Solve the inequation:
 $\left| \frac{3}{x-5} \right| > 1, x \neq 5$
 (A) (2, 3) (6, 7)
 (B) (2, 5) (6, 8)
 (C) (2, 3) (3, 5)
 (D) (2, 5) (5, 8)
- Q3** Find the number of integral values of ' x ' such that $|x-6| + |x-11| = 5$
- Q4** If $|3x + 5| = y + 14$ and $|y + 6| = 9$, find the maximum possible value of xy .
 (A) 30 (B) 12
 (C) 22 (D) 20
- Q5** The number of distinct real solutions of the equation $|x + 2| + |x - 2| = 6$ is:
- Q6** Find the sum of the distinct roots of the following equation.
 $|x^2 + x - 20| = x + 5$
- Q7** Find the number of possible integer values of a for which the quadratic equation $x^2 - 6x + |a - 4| = 0$ has two distinct real roots.
- Q8** The number of distinct possible solutions of the equation $|x|(16x^2 - 3) + 8x^2 = 0$ is:
 (A) 2 (B) 3
 (C) 4 (D) 5
- Q9** How many pairs of integers (x, y) satisfy the following equation?
 $(|x| - 4)(|y| + 8) = 32$
 (A) 10 (B) 4
 (C) 12 (D) 6
- Q10** The sum of all the distinct integer values of x satisfying the inequalities $(|x| - 4)(x + 9) < 0$ and $|x| < 9$ is:
 (A) 1 (B) 0
 (C) -1 (D) -9
- Q11** If x and y satisfy the equations $3|x| + 3x + 4y = 32$ and $3x + 2|y| - 2y = 22$, then $|x - y|$ equals:
- Q12** Find the sum of all integral values of a which satisfy the following inequality:
 $(|a + 2| - 5)(|a - 3| - 7) < 0$
- Q13** The number of integral values of ' x ' satisfying $|x - 16| + |36 - 2x| < 8$ is
- Q14** How many distinct ordered pairs (a, b) of integers are there such that $\frac{1}{|a|} + \frac{1}{|b|} = \frac{1}{3}$?
- Q15** Find the possible range of values of x if $|4x - 10| + |16 - 6x| = |6 - 2x|$.
 (A) $-\frac{7}{2} \leq x \leq \frac{5}{2}$
 (B) $\frac{5}{2} \leq x \leq \frac{8}{3}$
 (C) $-\frac{8}{3} \leq x \leq \frac{5}{2}$
 (D) $\frac{5}{2} < x < \frac{8}{3}$
- Q16** Find the number of integral values of x that satisfy $\frac{(|x-7|)(7-|x|)}{(|x|-1)(|x|^2-x+1)} > 0$.



- Q17** If $|x - 5| \geq 4$, $|4y + 7| \geq 13$ and $|z - 6| \geq 10$ and 'x', 'y' & 'z' are non-zero integers, then what is the minimum value of $|x| + |y| + |z|$?
- Q18** The number of distinct real values of x, satisfying the equation $\max\{x, y\} - \min\{x, y\} = |x + y| - |x - y|$, where $|y| \leq 1$, y being a non-negative integer, is:
- Q19** Find the number of real roots of the equation $x|x| - 7|x + 3| + 15 = 0$.
(A) 0 (B) 2
(C) 3 (D) 4
- Q20** The number of distinct pairs of integers (x, y) satisfying $|xy + 2| < |2x + y| < 4$ is:
- Q21** If $|m-4| \leq 3$, $|2n+5| \leq 11$ and $|o-5| \leq 9$ and 'm', 'n' & 'o' are non-zero integers, then what is the maximum value of $\frac{mn}{o}$?
- Q22** Determine the set of values that best describes x if $|5x - 11| + |15 - 7x| = |4 - 2x|$
(A) $-\frac{11}{5} \leq x \leq -\frac{15}{7}$
(B) $\frac{15}{7} \leq x \leq \frac{11}{5}$
(C) $-\frac{11}{5} < x < -\frac{15}{7}$
(D) $\frac{15}{7} < x < \frac{11}{5}$
- Q23** Given that $5 < x < 6$, where x is a real number, and $|x - m| + |x - n| = 5$, with 'm' and 'n' being real numbers such that $m \leq 5$ and $n \geq 6$, determine the product of the minimum value of 'm' and the maximum value of 'n'.
- Q24** If $|3(x-4)| + |x+6| \leq 15$, what will be the sum of all possible integer values of x?
(A) 9 (B) 17
(C) 12 (D) 14
- Q25** Determine the area (in square units) enclosed by the functions $f(x) = x - 1 + 4$ and $g(x) = 8 - x - 1$.
(A) 4 (B) 6

(C) 8 (D) 12

- Q26** Find the area bounded (in square units) by the region defined by inequalities $|x-2y| \leq 8$ and $|x+2y| \leq 8$.
- Q27** How many integral values exist for x if $|11x-5| \leq |11-5x|$
(A) 3 (B) 4
(C) 5 (D) 7
- Q28** Find the area enclosed by the graphs of $|x-1| + |y-2| \leq 7$ and $|x| \geq 3$ in square units.
- Q29** How many integral values of x satisfy the given inequality?
 $\frac{|x^2-4|}{|2x-1|} < 1$
- Q30** If $f(x) = 3x - 8$ and $g(x) = 5 - 3x$. Then for what values of x is the following equation satisfied $|f(x)| + |g(x)| = |f(x) + g(x)|$
(A) $\frac{5}{3} \leq x \leq \frac{8}{3}$
(B) $\frac{5}{3} < x < \frac{8}{3}$
(C) $x \leq \frac{5}{3}$ or $x \geq \frac{8}{3}$
(D) $x < \frac{5}{3}$ or $x > \frac{8}{3}$



Answer Key

Q1 D
Q2 D
Q3 6
Q4 B
Q5 2
Q6 3
Q7 17
Q8 B
Q9 A
Q10 B
Q11 7
Q12 28
Q13 5
Q14 12
Q15 B

Q16 10
Q17 7
Q18 3
Q19 C
Q20 4
Q21 56
Q22 B
Q23 10
Q24 D
Q25 C
Q26 64
Q27 A
Q28 34
Q29 3
Q30 A



Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Given that,

$$||x - 5| - 6| > 5$$

Case 1: When $|x - 5| - 6 > 5$

- $|x - 5| > 5 + 6$
- $|x - 5| > 11$
- $x - 5 < -11$ or, $x - 5 > 11$
- $x < -11 + 5$ or, $x > 11 + 5$
- $x < -6$ or, $x > 16$

Case 2: When $|x - 5| - 6 < -5$

- $|x - 5| < -5 + 6$
- $|x - 5| < 1$
- $-1 < x - 5 < 1$
- $-1 + 5 < x < 1 + 5$
- $4 < x < 6$

So, the required range is $(-\infty, -6)$ or $(4, 6)$ or $(16, \infty)$.

Video Solution:



Q2 Text Solution:

It is given that,

$$\left| \frac{3}{x-5} \right| > 1, x \neq 5$$

- $\frac{3}{x-5} < -1$ or $\frac{3}{x-5} > 1$
- $\frac{x-2}{x-5} < 0$ or $\frac{8-x}{x-5} > 0$

- $2 < x < 5$ or $5 < x < 8$, since $x \neq 5$.

The solution set of given inequation is $(2, 5) \cup (5, 8)$.

Video Solution:



Q3 Text Solution:

Here the critical points will be 6 and 11 and thus the minimum value of $|x-6| + |x-11|$ will be the distance between the two critical points 6 and 11 i.e. 5

This implies that for all the values of 'x' from 6 to 11 (both inclusive), $|x-6| + |x-11|$ will be equal to 5, before $x = 6$ and after $x = 11$, $|x-6| + |x-11|$ will keep increasing

So, the required number of such integral values will be 6 (6, 7, 8, 9, 10 and 11)

Video Solution:



Q4 Text Solution:

Given $|y + 6| = 9$,

$$y + 6 = \pm 9$$

$$y = 3 \text{ or } y = -15. \dots(1)$$

But, y cannot be -15, because if $y = -15$, then we get

$|3x + 5| = -15 + 14 = -1$, which is impossible as a modulus value cannot be negative.

$$\text{Also, } |3x + 5| = y + 14 \dots(2)$$

$$3x + 5 = y + 14 \text{ or } 3x + 5 = -y - 14$$



Case (i):

$$3x + 5 = y + 14$$

$$x = \frac{y+9}{3}$$

When $y = 3$, we get $x = 4$.

Therefore, the solution (x, y) is $(4, 3)$.

$$xy = 12$$

Case (ii):

$$3x + 5 = -y - 14$$

$$x = \frac{-(y+19)}{3}$$

When $y = 3$, we get $x = -\frac{22}{3}$.

Therefore, the solution (x, y) is $(-\frac{22}{3}, 3)$.

$$xy = -22$$

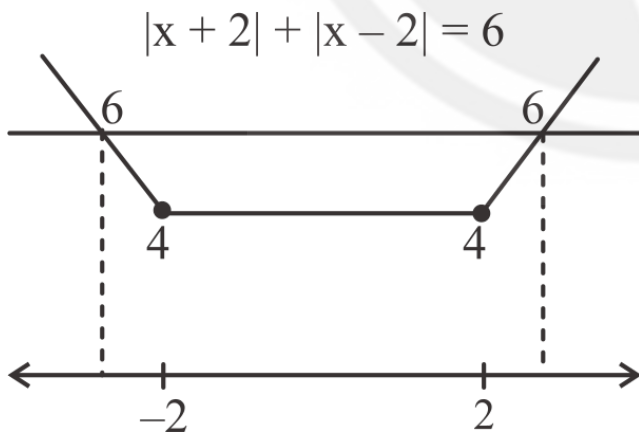
So, the maximum possible value of xy will be 12

Video Solution:



Q5 Text Solution:

The graph of the given function will look like below



Since at $x = -2$ and at $x = 2$, we have $|x+2| + |x-2| = 4$

Coefficient of 'x' is 2 which is positive, so the graph will keep increasing beyond those points. Now, we can see that $|x+2| + |x-2|$ will be = 6, for two distinct real values of 'x', one will be somewhere < -2 and the other would be > 2 .

Video Solution:



Q6 Text Solution:

Let $x^2 + x - 20 \geq 0$..(i), then we have

$$x^2 + x - 20 = x + 5$$

$$x^2 - 25 = 0$$

$$x = \pm 5$$

Since, $x = -5, 5$ both satisfy the inequality (i), so $x = -5, 5$ both are valid solutions.

Now, let $x^2 + x - 20 < 0$, then we have

$$-x^2 - x + 20 = x + 5$$

$$x^2 + 2x - 15 = 0$$

$$(x+5)(x-3) = 0$$

$$x = -5, 3$$

Since, $x = -5$ already satisfied the given inequality and also, $x = 3$, does the same, so both are valid solutions.

Hence, the required sum of the distinct roots = $-5 + 5 + 3 = 3$

Video Solution:



Q7 Text Solution:

The given quadratic equation is:

$$x^2 - 6x + |a-4| = 0 \dots (i)$$

The equation (i) will have two distinct real roots if

$$(-6)^2 - 4 \times 1 \times |a-4| > 0,$$

$$\text{i.e., } 36 - 4|a-4| > 0$$

$$\text{i.e., } 9 - |a-4| > 0 \dots (ii)$$

Case 1: When $a > 4$

Then, from (ii), we get



$$9 - (a - 4) > 0$$

$$9 - a + 4 > 0$$

$$a < 13$$

Therefore, $4 < a < 13$...(iii)

Case 2: When $a < 4$

Then, from (ii), we get

$$9 - (4 - a) > 0$$

$$9 - 4 + a > 0$$

$$a > -5$$

Therefore, $-5 < a < 4$...(iv)

Also, at $a = 4$, in eq. (i), we get

$$x^2 - 6x = 0$$

$$x(x - 6) = 0$$

$$x = 0, 6$$

So, $a = 4$ is also possible.

Hence, $-5 < a < 13$.

i.e., the number of possible integer values of a is 17.

Video Solution:



Q8 Text Solution:

The given equation is $|x|(16x^2 - 3) + 8x^2 = 0$

Let $y = |x|$.

Therefore, $y^2 = x^2$

So, we have

$$y(16y^2 - 3) + 8y^2 = 0$$

$$y(16y^2 + 8y - 3) = 0$$

$$y[16y^2 + 12y - 4y - 3] = 0$$

$$y[4y(4y + 3) - 1(4y + 3)] = 0$$

$$y(4y - 1)(4y + 3) = 0$$

So, the possible distinct solutions are $y = 0$ and

$\frac{1}{4}$. [Since y cannot be negative.]

But, since, $y = |x|$.

So, the total distinct possible solutions of the given equation are $0, -\frac{1}{4}, \frac{1}{4}$.

Thus, the equation has a total of 3 distinct possible solutions.

Video Solution:



Q9 Text Solution:

The given equation is $(|x| - 4)(|y| + 8) = 32$.

Since, x and y are integers, so $(|x| - 4)$ and $(|y| + 8)$ are also integers.

Now, 32 can be written as $32 \times 1, 2 \times 16$, or, 4×8 .

One thing can be noted that we have to eliminate the possibilities, where we have $(|y| + 8) < 0$ and where $(|y| + 8) < 8$ as $|y|$ is at least 0.

So, $(|y| + 8)$ can be 8, 16, or, 32. i.e., $|y|$ can be $8 - 8$, or, $16 - 8$, or, $32 - 8$ i.e., $|y|$ can be 0, or, 8, or, 24

So, when $|y| = 0$, then $(|x| - 4) = \frac{32}{8} = 4$

$|x| = 8$, i.e., $x = \pm 8$

i.e., we get 2 pairs of (x, y) from this case: $(-8, 0)$, and $(8, 0)$

Again, when $|y| = 8$, then $(|x| - 4) = \frac{32}{16} = 2$

$|x| = 6$, i.e., $x = \pm 6$

i.e., we get 4 pairs of (x, y) from this case: $(-6, -8)$, $(-6, 8)$, $(6, -8)$, and $(6, 8)$

Also, when $|y| = 24$, then $(|x| - 4) = \frac{32}{32} = 1$

$|x| = 5$, i.e., $x = \pm 5$. Similarly, we get 4 pairs of (x, y) from this case.

Therefore, we get total $= 2 + 4 + 4 = 10$ distinct pairs of (x, y) satisfying the given inequality.

Video Solution:



**Q10 Text Solution:**

The given inequality is $(|x| - 4)(x + 9) < 0$

Case 1: When $x \geq 0$

Then, we have

$$(x - 4)(x + 9) < 0$$

Using the Wavy-Curvy method, we have

$$(+)\quad -9\quad (-)\quad 4\quad (+)$$

$$\text{i.e., } -9 < x < 4$$

But, since, we have assumed $x \geq 0$, so the appropriate range of x in this case will be:

$$0 \leq x < 4 \dots (i)$$

Case 2: When $x < 0$

Then, we have

$$(-x - 4)(x + 9) < 0$$

$$(x + 4)(x + 9) > 0$$

Using the Wavy-Curvy method, we have

$$(+)\quad -9\quad (-)\quad -4\quad (+)$$

$$\text{i.e., } x \in (-\infty, -9) \cup (-4, \infty)$$

But, since, we have assumed that $x < 0$, the correct range of x in this case will be:

$$x \in (-\infty, -9) \cup (-4, 0) \dots (ii)$$

Thus, by combining (i), and (ii) the range of the inequality is:

$$(-\infty, -9) \cup (-4, 4).$$

Again, since $|x| < 9$, so we can write the revised range as:

$$x \in (-4, 4)$$

Hence, the required sum = $-3 - 2 - 1 + 0 + 1 + 2 + 3 = 0$

Video Solution:

**Q11 Text Solution:**

Here, we have been given two equations

$$3|x| + 3x + 4y = 32 \dots (i)$$

$$3x + 2|y| - 2y = 22 \dots (ii)$$

Considering Equation (ii), if

$$y \geq 0, \text{ then } x = \frac{22}{3}, \text{ but then,}$$

Equation (i) will never get satisfied, and thus, $y < 0$.

Therefore from Equation (ii), we have,

$$3x - 4y = 22 \dots (iii)$$

Considering Equation (i), if

$x < 0$, then $y = 8$, which is not possible as y has to be less than 0.

Therefore, we will have

$$x \geq 0,$$

Therefore from Equation (i), we have,

$$6x + 4y = 32 \dots (iv)$$

Upon solving Equations (iii) and (iv), we get

$$x = 6 \text{ and } y = -1,$$

$$\text{Thus, } |x - y|$$

$$= |6 - (-1)|$$

$$= 7$$

Video Solution:

**Q12 Text Solution:**

The given inequality is:

$$(|a + 2| - 5)(|a - 3| - 7) < 0$$

Case I: When $a < -2$. Then, we have

$$(-a - 2 - 5)(-a + 3 - 7) < 0$$

$$(a + 7)(a + 4) < 0$$

Using the Wavy-Curvy Method, we get

$$(+)\quad -7\quad (-)\quad -4\quad (+)$$



i.e., $a \in (-7, -4)$. So, the possible integer values of a are: **- 6 and - 5.**

Case II: When $-2 \leq a < 3$. Then, we get

$$(a + 2 - 5)(-a + 3 - 7) < 0$$

$$(a - 3)(a + 4) > 0$$

Using the Wavy-Curvy Method, we get

$$\begin{array}{ccccccc} (+) & & -4 & & (-) & & 3 & & (+) \end{array}$$

i.e., $a \in (-\infty, -4) \cup (3, \infty)$. Since, $-2 \leq a < 3$, and all the solutions are outside the range, so we can eliminate this case.

Case III: When $a \geq 3$. Then, we have

$$(a + 2 - 5)(a - 3 - 7) < 0$$

$$(a - 3)(a - 10) < 0$$

Using the Wavy-Curvy Method, we get

$$\begin{array}{ccccccc} (+) & & 3 & & (-) & & 10 & & (+) \end{array}$$

i.e., $a \in (3, 10)$, i.e., So, the possible integer values of a are: **4, 5, 6, 7, 8 and 9.**

Thus, the required sum of all the possible integer values of a

$$\begin{aligned} &= -6 - 5 + 4 + 5 + 6 + 7 + 8 + 9 \\ &= 28 \end{aligned}$$

Video Solution:



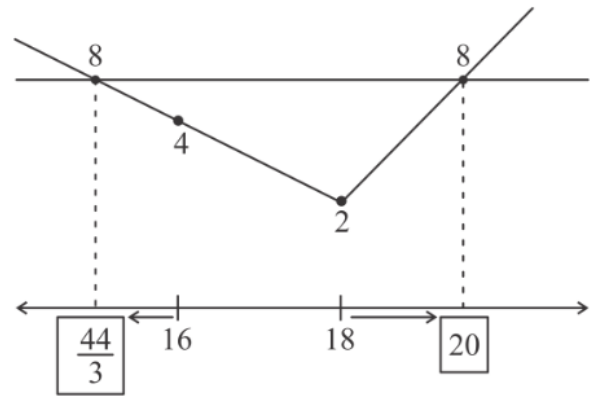
Q13 Text Solution:

We can write $|x - 16| + |36 - 2x| < 8$ as

$$|x - 16| + |2x - 36| < 8$$

$$|x - 16| + 2|x - 18| < 8$$

Thus, the critical points here will be 16 and 18



At $x = 16$, the value of the given expression is 4 and at $x = 18$, its value is 2

Also, the coefficient of 'x' in $x + 2x = 3x$ is positive, so the curve will keep increasing beyond these critical points

Thus, we can see that it will be equal to 8 for two values of 'x', one of which will be lesser than 16 and one greater than 18.

At 18, it is equal to 2, for it to be equal to 8, we need an increase of 6, this increase needs to be divided by the coefficient of 'x' i.e. 3 which gives us 2

So, the required value of 'x' at which the given expression is equal to 8 is $18 + 2 = 20$

Similarly, for $x = 16$, value of the expression is 4, for it to be equal to 8, we need an increase of 4, dividing by the coefficient of 'x', we get $\frac{4}{3}$

So, the other value of 'x' which the given expression is equal to 8 is $16 - \frac{4}{3}$ (14.67 approx.)

Thus, we can say that for all the values of 'x' between 14.67 and 20, the expression will be less than 8

Therefore, the required number of such integral values of 'x' is **5** (from 15 to 19)

Video Solution:



Q14 Text Solution:



[Android App](#)

[iOS App](#)

[PW Website](#)

Given that,

$$\frac{1}{|a|} + \frac{1}{|b|} = \frac{1}{3}$$

Let $|a| = x$ and $|b| = y$. Therefore, we have

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{3}$$

$$3x + 3y = xy$$

$$xy - 3x - 3y + 9 = 9$$

$$x(y - 3) - 3(y - 3) = 9$$

$$(x - 3)(y - 3) = 9$$

Let's explore all the possibilities here:

9 as a product of two integers	value of (x-3)	value of (y-3)	values of x and y
1×9	1	9	$x=4, y=12$
9×1	9	1	$x=12, y=4$
3×3	3	3	$x=6, y=6$
$(-1) \times (-9)$	-1	-9	$x=2, y=-6$
$(-9) \times (-1)$	-9	-1	$x=-6, y=2$
$(-3) \times (-3)$	-3	-3	$x=0, y=0$ (Invalid Solution)

Thus, 5 valid ordered pairs of integers of (x, y) are: (4, 12), (12, 4), (6, 6), (2, -6), and (-6, 2).

But, since, $x = |a|$ and $y = |b|$, therefore, the x, y cannot be negative. So, the actual ordered pairs of ($|a|$, $|b|$) which will satisfy the inequality are:

(4, 12), (12, 4), and (6, 6).

Thus, a total of $3 \times 4 = 12$ different ordered pairs (a, b) of integers will satisfy the given inequality.

[Since, for e.g., if ($|a|$, $|b|$) = (4, 12), then (a, b) = (4, 12), (-4, 12), (4, -12), and (-4, -12).

Similarly, we will get 4 pairs of (a, b) for each of the other two pairs of ($|a|$, $|b|$).]

Video Solution:



Q15 Text Solution:

The given equation is:

$$|4x - 10| + |16 - 6x| = |6 - 2x|$$

Let's rewrite the given equation as:

$$|4x - 10| + |6x - 16| - |2x - 6| = 0 \dots(i)$$

$$\text{Let } y = |4x - 10| + |6x - 16| - |2x - 6| \dots(ii)$$

The critical points are: $x = \frac{5}{2}, \frac{8}{3}$ and 3.

Let's put the critical points on a number line. Now, let's find the values of y at these critical points.

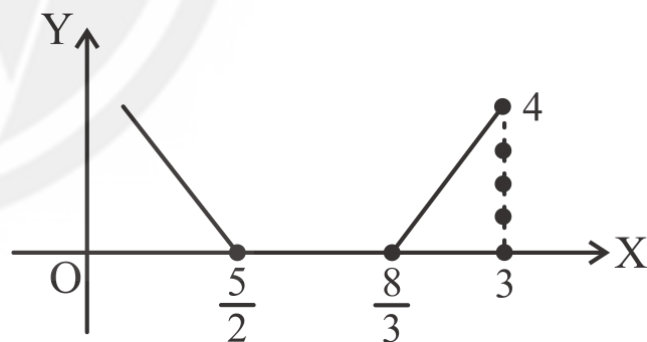
$$\text{At } x = 3, y = |2| + |2| - 0 = 4$$

$$\text{At } x = \frac{5}{2}, y = 0 + 1 - 1 = 0$$

$$\text{At } x = \frac{8}{3}, y = \frac{2}{3} + 0 - \frac{2}{3} = 0$$

So, let's plot these data on the number line.

Also, let's find the sum of the coefficients of x in each expression inside the modulus = $4 + 6 - 2 = 8 > 0$. So, the curve will look like as follows:



Hence, the range of x where the y will be zero is:

$\frac{5}{2} \leq x \leq \frac{8}{3}$, which is the required range of x satisfying the given inequality.

Video Solution:





Q16 Text Solution:

The given inequality is:

$$\frac{(|x-7|)(7-|x|)}{(|x|-1)(|x|^2-x+1)} > 0$$

Note that, irrespective of the values of x , $|x-7|$ is always non-negative

Again, for all real values of $|x|^2 = x^2$. Also, the leading coefficient of $x^2 - x + 1$ is positive and the discriminant $(-1)^2 - 4(1)(1) = -3 < 0$, so the curve of the graph never touches the X-axis. Thus, $x^2 - x + 1$, i.e., $|x|^2 - x + 1 > 0$ for all values of x .

So, our inequality reduces to:

$$\frac{(7-|x|)}{(|x|-1)} > 0$$

Now, the critical points are: -7, -1, 1, and 7.

Case (i): When $x < -7$, then the numerator is negative, but the denominator is positive, and thus the inequality is not satisfied.

Case (ii): When $-7 < x < -1$, the numerator and denominator both are positive, so the inequality **is satisfied**.

Case (iii): When $-1 < x < 1$, the numerator is positive, but the denominator is negative, and thus the inequality is not satisfied.

Case (iv): When $1 < x < 7$, the numerator and denominator both are positive, so the inequality **is satisfied**.

Case (v): When $x > 7$, the numerator is negative, but the denominator is positive, so the inequality is not satisfied.

Hence, the solution set for the inequality is:

$$x \in (-7, -1) \cup (1, 7).$$

i.e., total of $5 + 5 = 10$ integer values of x satisfies the given inequality.

Video Solution:



Q17 Text Solution:

$$|x-5| \geq 4$$

$$\Rightarrow x-5 \geq 4 \quad \text{or} \quad 5-x \geq 4$$

$$\Rightarrow x \leq 1 \quad \text{or} \quad x \geq 9$$

$$|4y+7| \geq 13$$

$$\Rightarrow -4y-7 \geq 13 \quad \text{or} \quad 4y+7 \geq 13$$

$$\Rightarrow 4y \leq -20 \quad \text{or} \quad 4y \geq 6$$

$$\Rightarrow y \leq -5 \quad \text{or} \quad y \geq 1.5$$

$$|z-6| \geq 10$$

$$\Rightarrow 6-z \geq 10 \quad \text{or} \quad z-6 \geq 10$$

$$\Rightarrow z \leq -4 \quad \text{or} \quad z \geq 16$$

So, the minimum value of $|x| + |y| + |z| = 1 + 2 + 4 = 7$

[Since, x , y , and z are non-zero integers. Also, we had to take the smallest possible numerical values of x , y and z to obtain the minimum of $|x| + |y| + |z|$.]

Video Solution:



Q18 Text Solution:

Given the equation:

$$\max\{x, y\} - \min\{x, y\} = |x+y| - |x-y|$$

where $|y| \leq 1$ and y is a non-negative integer, we analyze each possible y :

Case 1: $y = 0$

$$|x| = |x| - |x| \implies |x| = 0 \implies x = 0$$

Solution for $y = 0$: $x = 0$.



Case 2: $y = 1$

$$|x - 1| = |x + 1| - |x - 1| \implies 2|x - 1| = |x + 1|$$

Subcase (a): $x < -1$

$$2(1 - x) = -x - 1 \implies x = 3$$

Subcase (b): $-1 \leq x < 1$

$$2(1 - x) = x + 1 \implies x = \frac{1}{3}$$

Subcase (c): $x \geq 1$

$$2(x - 1) = x + 1 \implies x = 3$$

Solutions for $y = 1$: $x = \frac{1}{3}, 3$

Combining all valid solutions:

$$x = 0, \frac{1}{3}, 3$$

The number of distinct real values of x : 3.

Video Solution:



Q19 Text Solution:

Case 1: When $x < -3$

Then, we have

$$-x^2 + 7(x + 3) + 15 = 0$$

$$-x^2 + 7x + 36 = 0$$

$$x^2 - 7x - 36 = 0$$

$$x = \frac{7 \pm \sqrt{193}}{2}$$

So, x only can be $\frac{7 - \sqrt{193}}{2}$.

Case 2: When $-3 \leq x < 0$

Then, we have

$$-x^2 - 7(x + 3) + 15 = 0$$

$$-x^2 - 7x - 6 = 0$$

$$x^2 + 7x + 6 = 0$$

$$(x + 1)(x + 6) = 0$$

So, x only can be -1 .

Case 3: When $x \geq 0$

Then, we have

$$x^2 - 7(x + 3) + 15 = 0$$

$$x^2 - 7x - 6 = 0$$

$$x = \frac{7 \pm \sqrt{73}}{2}$$

So, x only can be $\frac{7 + \sqrt{73}}{2}$.

Hence, a total of 3 real values of x are possible.

Video Solution:



Q20 Text Solution:

Given that,

$$|xy + 2| < |2x + y| \dots (i)$$

$$4 + (xy)^2 + 4xy < 4x^2 + y^2 + 4xy$$

$$-4x^2 - y^2 + 4 + (xy)^2 < 0$$

$$4(1 - x^2) - y^2(1 - x^2) < 0$$

$$(4 - y^2)(1 - x^2) < 0$$

Case 1: $(4 - y^2) < 0$ and $(1 - x^2) > 0$
i.e., $y < -2$ or $y > 2$ and $x = 0$

Case 2: $(4 - y^2) > 0$ and $(1 - x^2) < 0$
i.e., $y = 0, \pm 1$ and $x < -1$ or $x > 1$

Now, take $|2x + y| < 4 \dots (ii)$

For Case 1, when $x = 0$, we have

$$|y| < 4, \text{ i.e., } -4 < y < 4$$

So, in this case, the possible integer pairs of (x, y) are **(0, -3), (0, 3)**.

For Case 2,

When $y = 0$, we have

$$|2x| < 4 \text{ i.e., } |x| < 2 \text{ i.e., } -2 < x < 2$$

So, in this case, we will not get any possible integer pairs of (x, y) .

When $y = -1$, we have

$$|2x - 1| < 4$$

$$-4 < 2x - 1 < 4$$

$$-3 < 2x < 5$$

$$-1.5 < x < 2.5$$



So, in this case, the possible integer pair of (x, y) is **(2, -1)**.

When $y = 1$, we have

$$|2x + 1| < 4$$

$$-4 < 2x + 1 < 4$$

$$-5 < 2x < 3$$

$$-2.5 < x < 1.5$$

So, in this case, the possible integer pair of (x, y) is **(-2, 1)**.

Hence, the total number of possible distinct integer pairs (x, y) is 4.

Video Solution:



Q21 Text Solution:

$$|m - 4| \leq 3$$

$$\Rightarrow 4 - m \leq 3 \text{ or } m - 4 \leq 3$$

$$\Rightarrow 1 \leq m \leq 7$$

$$|2n + 5| \leq 11$$

$$\Rightarrow -2n - 5 \leq 11 \text{ or } 2n + 5 \leq 11$$

$$\Rightarrow -16 \leq 2n \leq 6$$

$$\Rightarrow -8 \leq n \leq 3$$

$$|o - 5| \leq 9$$

$$\Rightarrow 5 - o \leq 9 \text{ or } o - 5 \leq 9$$

$$\Rightarrow -4 \leq o \leq 14$$

To maximize $\frac{mn}{o}$, generally we should maximize numerator and minimize the denominator however since negative values are also being considered, to obtain the maximum value, if the highest absolute value of the numerator is positive then we should consider minimum absolute positive value of the denominator, whereas if the highest absolute value of the numerator is negative then the numerator should be the minimum absolute negative value of the denominator. For example if we would consider only the maximum value of m and n we will obtain mn

as 21 whereas maximum absolute value that can be obtained here is $-8 \times 7 = -56$ And hence to minimize the denominator and to maximize the fraction the minimum absolute negative value of o, i.e., -1 will be considered, so

$$\text{Maximum value of } \frac{mn}{o} = \frac{-56}{-1} = 56$$

Video Solution:



Q22 Text Solution:

$$|5x - 11| + |15 - 7x| = |4 - 2x|$$

However

Simply

$$(5x - 11) + (15 - 7x) = (4 - 2x)$$

So if

$$y = 5x - 11 \text{ and } z = 15 - 7x, \text{ so}$$

$$|y| + |z| = |y + z|$$

Which means if $y < 0$, $z < 0$ and if $y > 0$, $z > 0$. So

Case 1 : $y < 0$ and $z < 0$

$$5x - 11 < 0 \text{ and } 15 - 7x < 0$$

$$\Rightarrow x < \frac{11}{5} \text{ and } x > \frac{15}{7}$$

So

$$\frac{15}{7} < x < \frac{11}{5}$$

Case 2 : $y > 0$ and $z > 0$

$$5x - 11 > 0 \text{ and } 15 - 7x > 0$$

$$\Rightarrow x > \frac{11}{5} \text{ and } x < \frac{15}{7}$$

which is not possible.

Checking for the critical points, we see that both $15/7$ and $11/5$ also satisfy, so

$$\frac{15}{7} \leq x \leq \frac{11}{5}$$

Video Solution:



**Q23 Text Solution:**

Since $5 < x < 6$ and $m \leq 5$ and $n \geq 6$

$$|x - m| + |x - n| = 5$$

$$\Rightarrow x - m + n - x = 5$$

$$\Rightarrow n - m = 5$$

So n will be maximum when m will be minimum and m will be minimum when n will be maximum.

So

$$n_{\min} = 6 \text{ and } m_{\max} = 5$$

So

$$m_{\min} = n_{\min} - 5 = 6 - 5 = 1$$

$$n_{\max} = m_{\max} + 5 = 5 + 5 = 10$$

So

$$m_{\min} \times n_{\max} = 1 \times 10 = 10$$

Video Solution:

**Q24 Text Solution:**

The deflection points will be $x=4$ and $x=-6$, so

For $x > 4$

$$3(x - 4) + (x + 6) \leq 15$$

$$\Rightarrow 4x - 6 \leq 15$$

$$\Rightarrow x \leq \frac{21}{4}$$

So possible integral value of x in the given range is 5

For $-6 \leq x \leq 4$

$$3(4 - x) + (x + 6) \leq 15$$

$$\Rightarrow 18 - 2x \leq 15$$

$$\Rightarrow 2x \geq 3$$

$$\Rightarrow x \geq \frac{3}{2}$$

So possible integral values of x here are 2, 3 and 4

For $x < -6$

$$3(4 - x) - (x + 6) \leq 15$$

$$\Rightarrow 6 - 4x \leq 15$$

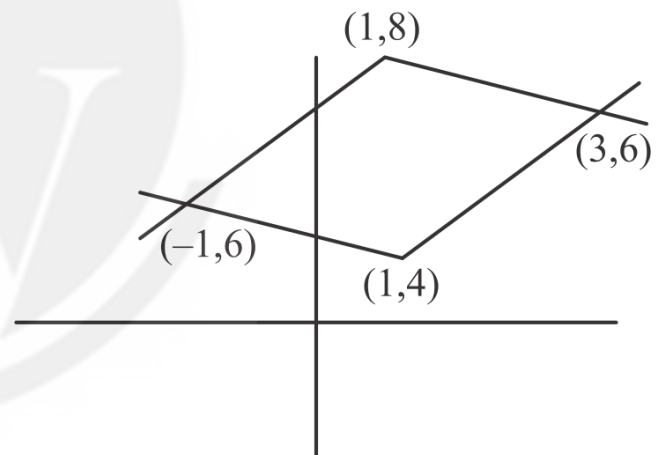
$$\Rightarrow 4x \geq -9$$

$$\Rightarrow x \geq -\frac{9}{4}$$

So no possible integral value of x exists in this range.

So the possible integral values of x are 2, 3, 4 and 5 and the sum of these values is 14.

Video Solution:

**Q25 Text Solution:**

$$f(x) = |x-1| + 4$$

For $x > 1$

$$f(x) = x - 1 + 4 = x + 3$$

For $x < 1$

$$f(x) = 1 - x + 4 = 5 - x$$

For $x = 1$

$$f(x) = 4$$

Similarly

$$g(x) = 8 - |x-1|$$

For $x > 1$

$$g(x) = 8 - x - 1 = 9 - x$$

For $x < 1$



$$g(x) = 8 - 1 + x = x + 7$$

For $x=1$

$$g(x) = 8$$

For $x < 1$, $f(x)$ and $g(x)$ will intersect at

$$5 - x = x + 7$$

$$\Rightarrow x = -1$$

At $x = -1$

$$f(x) = g(x) = 5 - x = x + 7 = 6$$

For $x > 1$, $f(x)$ and $g(x)$ will intersect at

$$x + 3 = 9 - x$$

$$\Rightarrow x = 3$$

At $x = 3$

$$f(x) = g(x) = x + 3 = 9 - x = 6$$

So the shape formed (lets call it ABCD) will have four vertices $A(1,4)$, $B(3,6)$, $C(1,8)$ and $D(-1,6)$.

$$AB = \sqrt{(3-1)^2 + (6-4)^2} = 2\sqrt{2} \text{ units}$$

$$BC = \sqrt{(1-3)^2 + (8-6)^2} = 2\sqrt{2} \text{ units}$$

$$CD = \sqrt{(-1-1)^2 + (6-8)^2} = 2\sqrt{2} \text{ units}$$

$$AD = \sqrt{(-1-1)^2 + (6-4)^2} = 2\sqrt{2} \text{ units}$$

$$AC = \sqrt{(1-1)^2 + (8-4)^2} = 4 \text{ units}$$

$$BD = \sqrt{(-1-3)^2 + (6-6)^2} = 4 \text{ units}$$

So all the four sides are equal and both the diagonals are equal. So it is a square.

$$\text{Ar}(ABCD) = (2\sqrt{2})^2 = 8 \text{ square units}$$

Video Solution:



Q26 Text Solution:

So the lines corresponding to the inequalities are

For

$$|x - 2y| \leq 8$$

$$x - 2y = 8$$

$$\text{and } x - 2y = -8$$

For

$$|x + 2y| \leq 8$$

$$x + 2y = 8$$

$$\text{and } x + 2y = -8$$

$x - 2y = 8$ and $x - 2y = -8$ will be parallel and $x + 2y = 8$ and $x + 2y = -8$ will be parallel.

$x - 2y = 8$ and $x + 2y = 8$ will intersect at

$$x - 2y + x + 2y = 8 + 8$$

$$\Rightarrow 2x = 16$$

$$\Rightarrow x = 8$$

$$x + 2y = 8$$

$$\Rightarrow 8 + 2y = 8$$

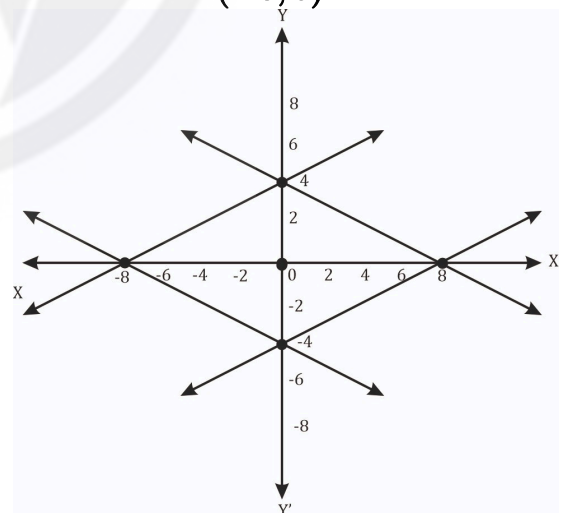
$$\Rightarrow y = 0$$

So they will intersect at $(8, 0)$.

Similarly $x - 2y = 8$ and $x + 2y = -8$ will intersect at $(0, -4)$.

$x - 2y = -8$ and $x + 2y = 8$ will intersect at $(0, 4)$.

and $x - 2y = -8$ and $x + 2y = -8$ will intersect at $(-8, 0)$.



So they form a rhombus.

Length of diagonal 1 = 8 to 8 so 16 units

Length of diagonal 2 = 4 to 4 so 8 units

So area of the rhombus =

$$\frac{1}{2} \times 16 \times 8 = 64 \text{ unit}^2$$



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Video Solution:**Q27 Text Solution:**

Critical points will be

$$11x-5=0 \text{ and } 11-5x=0$$

$$\Rightarrow x = \frac{5}{11} \text{ and } x = \frac{11}{5}$$

So checking

$$\text{For } x \geq \frac{11}{5}$$

$$11x - 5 \leq 5x - 11$$

$$\Rightarrow 6x \leq -6$$

$$\Rightarrow x \leq -1$$

But since $x \geq \frac{11}{5}$, no value will satisfy this case

$$\text{For } \frac{5}{11} \leq x < \frac{11}{5}$$

$$11x - 5 \leq 11 - 5x$$

$$\Rightarrow 16x \leq 16$$

$$\Rightarrow x \leq 1$$

So one integral value in this case will satisfy the inequality, $x=1$

$$\text{For } x < \frac{5}{11}$$

$$5 - 11x \leq 11 - 5x$$

$$\Rightarrow -6x \leq 6$$

$$\Rightarrow x \geq -1$$

So two integral values in this case will satisfy the inequality, $x=0$ and $x=-1$

So total three integral values.

Video Solution:**Q28 Text Solution:**

Lets find the vertices of the quadrilateral formed by $|x-1| + |y-2| \leq 7$.

Taking $|y-2| = 0$, i.e, $y=2$

$$|x-1| = 7$$

$$\Rightarrow x-1=7 \text{ or } 1-x=7$$

$$\Rightarrow x=8 \text{ or } -6$$

So two of the vertices are $(8,2)$ and $(-6,2)$

Taking $|x-1| = 0$, i.e., $x=1$

$$|y-2| = 7$$

$$\Rightarrow y-2=7 \text{ or } 2-y=7$$

$$\Rightarrow y=9 \text{ or } y=-5$$

So the other two vertices are $(1,9)$ and $(1,-5)$

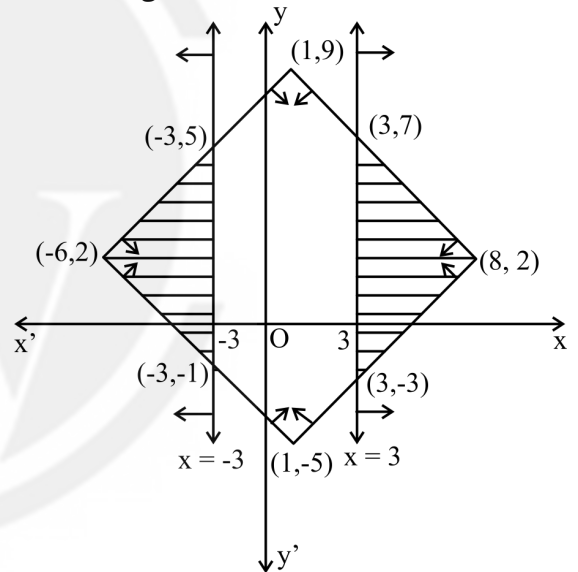
Also the region covered in the smaller than cases of such modulus equations lies inside the quadrilateral formed.

For $|x| \geq 3$

$$x \geq 3$$

$$\text{or } x \leq -3$$

So we will obtain the following graph by combining these two



Two triangles have been formed for the area enclosed within these two graphs

Lets find the vertices of their bases.

$x=3$ and $|x-1| + |y-2| = 7$ will intersect each other at

$$|3-1| + |y-2| = 7$$

$$\Rightarrow |y-2| = 5$$

$$\Rightarrow y-2=5 \text{ or } 2-y=5$$

$$\Rightarrow y=7 \text{ or } y=-3$$

So these two vertices are $(3,7)$ and $(3,-3)$

$x=-3$ and $|x-1| + |y-2| = 7$ will intersect each other at



$$|-3-1| + |y-2| = 7$$

$$|y-2| = 3$$

$$\Rightarrow y-2 = 3 \text{ or } 2-y = 3$$

$$\Rightarrow y = 5 \text{ or } y = -1$$

So these two vertices are $(-3, 5)$ and $(-3, -1)$

So by observation, the heights and bases of these two triangles are 3 units and 6 units for first triangle and 5 units and 10 units respectively.

So area bounded by the graphs =

$$\frac{1}{2} \times 3 \times 6 + \frac{1}{2} \times 5 \times 10 = 9 + 25 = 34 \text{ units}^2$$

Video Solution:



Q29 Text Solution:

$$\frac{|x^2-4|}{|2x-1|} < 1$$

$$\Rightarrow \left| (x+2)(x-2) \right| < |2x-1|$$

Hence the critical points here are

$$x=2, x=\frac{1}{2} \text{ and } x=-2$$

For all the ranges given below, we need to evaluate each of the expressions according to whether they will be positive or negative in the given range.

Range of x	Solution	Acceptable values of x
$x \geq 2$	$x^2 - 4 < 2x - 1$ $\Rightarrow x^2 - 2x - 3 < 0$ $\Rightarrow (x-3)(x+1) < 0$ $\Rightarrow -1 < x < 3$	$x=2$

$\frac{1}{2} < x < 2$	$4 - x^2 < 2x - 1$ $\Rightarrow x^2 + 2x - 5 > 0$ $\Rightarrow x < -1 - \sqrt{6} \text{ or } x > -1 + \sqrt{6}$ (Since, $2 < \sqrt{6} < 3$, so considering only integral values) $x < -3 \text{ or } x > 1$ (because $-1 - \sqrt{6}$ will be $-3.xx$ and $-1 + \sqrt{6}$ will be $1.xx$)	No acceptable value
$-2 \leq x < \frac{1}{2}$	$4 - x^2 < 1 - 2x$ $\Rightarrow x^2 - 2x - 3 > 0$ $\Rightarrow x < -1 \text{ or } x > 3$	$x=-2$
$x < -2$	$x^2 - 4 < 1 - 2x$ $\Rightarrow x^2 + 2x - 5 < 0$ $\Rightarrow -1 - \sqrt{6} < x < -1 + \sqrt{6}$ (Since, $2 < \sqrt{6} < 3$, so considering only integral values) $-4 < x < 2 \text{ (because } -1 - \sqrt{6} \text{ will be } -3.xx \text{ and } -1 + \sqrt{6} \text{ will be } 1.xx)$	$x=-3$

So there are 3 integral values of x.

Video Solution:



Q30 Text Solution:

Since

$$|f(x)+g(x)| = |f(x)| + |g(x)|$$

This means

$$\text{If } f(x) \leq 0 \text{ then } g(x) \leq 0 \text{ (both negative)}$$



and if $f(x) \geq 0$ then $g(x) \geq 0$ (both positive)
(Both needs to be of the same nature as
absolute value will be same if both have same
sign)

So

$$f(x) \times g(x) \geq 0$$

$$\Rightarrow (3x - 8)(5 - 3x) \geq 0$$

$$\Rightarrow (3x - 8)(3x - 5) \leq 0$$

$$\Rightarrow x \leq \frac{8}{3} \text{ or } x \geq \frac{5}{3}$$

$$\Rightarrow \frac{5}{3} \leq x \leq \frac{8}{3}$$

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